

Regular complete-packet ownership and H_N/REG closure

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope: corrected conclusion

R-102 closed the last analytic term in the regular rational current,

$$\mathcal{K}_R = G^T Lc + \frac{1}{2}c^T B_1 c, \quad B_1 = (P_{\Sigma_{\text{tar}}} B_R)(U + a), \quad (1.1)$$

uniformly in the terminal cutoff and deterministic positive-semidefinite target and future heat. Its proof leaves the full heat-lifted terminal square in (1.1) unspent. The open question after R-102 was not another multiplier estimate. It was whether the accepted low, FAR, linear, raw-Wick, unshifted-current, shifted-current, payment, heat, trace, conditional-mean, and forest authorities really reconstruct the complete regular packet with every owner used exactly once.

The two global budgets used throughout are, pathwise,

$$x_k = \|h_k\|_{\mathcal{H}_k}^2, \quad X = \sum_{k>\ell} x_k, \quad Y = \|U + A^*\|_6^6. \quad (1.1a)$$

Thus X is the accepted high-shell Cameron–Martin energy, not a terminal H^2 norm, and Y is the terminal sextic stabilizer.

This note proves that reconstruction. There are two conclusions.

1. The complete signed linear–rational NEAR packet on the R-079 regular mutually orthogonal strict-past no-revisit class satisfies, for every $\eta, \zeta > 0$,

$$\boxed{\mathbb{E} \mathfrak{H}_N^{\text{reg}} \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta, \zeta, e, j_0}.} \quad (1.2)$$

2. Adding R-083 polynomial FAR and R-092 regular projected Cartan FAR closes the regular packet architecture REG. With the production action coefficients

$$c_X = \frac{9}{20}, \quad c_Y = \frac{3}{20}, \quad (1.3)$$

one may choose the conservative downstream-compatible allocations $\eta_* = 1/440$ and $\zeta_* = 3/100$ and obtain

$$\boxed{\mathbb{E} [\mathfrak{P}_{\text{reg}} + c_X X + c_Y Y] \geq (c_X - \eta_*) \mathbb{E} X + (c_Y - \zeta_*) \mathbb{E} Y - C \geq -C.} \quad (1.4)$$

The constant is uniform in the terminal cutoff.

The source coefficient is $c_X = 1/(2q)$ at $q = 10/9$. For compatibility with the optional R-088 comparison coordinate, put $p = 11/10$ and $\epsilon_v = 9/20$. Then

$$\eta_* = \frac{1}{2} \left(\frac{1}{2p} - \epsilon_v \right) = \frac{1}{440}, \quad \zeta_* = \frac{c_Y}{5} = \frac{3}{100}. \quad (1.5)$$

Thus the eight-module allocation uses $1/3520$ and $3/800$ per module. The R-088 number $1/220$ is only the strict threshold for an *additional* comparison charge, not a universal cap on internal allocations. Here it is merely a conservative benchmark. The proof itself only needs positive finite allocations strictly below the full action coefficients in (1.3).

The scope is deliberately unchanged: finite cutoff, fixed positive floor, the A1 production frame, common real-even local regulators, deterministic positive-semidefinite heat increments, the fixed R-096 collar, and annular mutually orthogonal strict-past no-revisit physical-shell controls with the accepted pathwise prefix ledgers. No arbitrary progressive/revisit theorem, $\text{OVERLAP}_{\text{src}}$ bound, Nelson estimate, cutoff or floor removal, interacting measure, or Sector-A closure is claimed.

The standalone R-085 averaged Cartan atom (4.11) is not proved here. R-092 provides a different integrated regular Cartan theorem. Likewise the old R-085 rational form (6.5) is not proved term by term: R-101 showed that such a proof would duplicate its raw-Wick owner, replaced the true frontier by (1.1), and R-102 closed that smaller frontier. These two old standalone statements are superseded as necessary premises for regular REG, not promoted to theorems.

2 Canonical packet and exact spatial partition

Put $\ell = j_0 - 1$, $A^* = A^{(J)}$, $A^0 = A^{(\ell)}$, and write the complete R-079 current as

$$\mathcal{J}(H)_{\alpha i} = Q_{\text{II}}^{1/2} M_{\alpha}(U + H)^T (D_i U + D_i H). \quad (2.1)$$

The covariance-normal endpoint is

$$\mathcal{W}_J(H) = \frac{1}{2} \|\mathcal{J}(H)\|_{\mathcal{H}}^2 - \frac{1}{2} \sum_i \int B(U + H) : \Gamma_{J,i} dx. \quad (2.2)$$

R-079 Theorem 3.1 and equation (4.4) give the exact safe-packet identity

$$\mathbb{E}\mathfrak{B}_{J,\text{safe}}^> = \mathfrak{L}_{\ell} + \sum_{j=j_0}^J (\mathfrak{F}_j + \mathfrak{J}_j) - \mathbb{E}\{\mathcal{R}_{\leq}(A^*) - \mathcal{R}_{\leq}(A^0)\}, \quad (2.3)$$

and its endpoint reconstruction

$$\mathbb{E}\mathfrak{B}_{J,\text{alg}}^>(A^*) = \mathbb{E}\mathfrak{B}_{J,\text{alg}}^>(A^0) + \mathbb{E}\mathfrak{B}_{J,\text{safe}}^>. \quad (2.4)$$

The first term on the right of (2.4) is the complete low safe endpoint,

$$\mathfrak{B}_{\text{low}} = \Delta V_J^{\text{ren}}(U; A^0) - N_{3,\text{nr}}(U; A^0) - \widehat{T}_{\leq}(U; A^0), \quad (2.5)$$

and is different from the conditional-low current $\mathfrak{L}_{\ell}$ in (2.3).

Apply the exact R-083 physical-shell FAR/NEAR projections to (2.3), before any estimate. They are a partition of the already reconstructed endpoint. The three controlled polynomial stopped-FAR rows vanish for the declared collar. The remaining FAR term is the projected Cartan block. The NEAR term is the direct sum of the three linear Pauli–Fierz rows and the single rational row. No coefficient, root, heat, trace, or forest term is created by this partition.

Write $\langle Z \rangle = \mathbb{E}Z$. Let $\mathfrak{C}_{\text{far}}^{\text{reg}}$, $\Delta\mathcal{W}_{L,\Sigma}$, and $\Delta\mathcal{W}_R$ denote the signed random Cartan, complete-linear, and rational blocks after the R-096 complete-cancellation-first ordering. Define the one signed paid difference by

$$\mathfrak{P}_{\text{paid}} := -\{\mathcal{R}_{\leq}(A^*) - \mathcal{R}_{\leq}(A^0)\}, \quad \mathcal{R}_{\leq}(H) = N_{3,\text{nr}}(H) + \widehat{T}_{\leq}(H). \quad (2.5a)$$

Then the definitions give the exact expectation-level decomposition

$$\boxed{\langle \mathfrak{P}_{\text{reg}} \rangle = \langle \mathfrak{B}_{\text{low}} \rangle + \mathfrak{L}_{\ell} + \langle \mathfrak{C}_{\text{far}}^{\text{reg}} \rangle + \langle \Delta\mathcal{W}_{L,\Sigma} \rangle + \langle \Delta\mathcal{W}_R \rangle + \langle \mathfrak{P}_{\text{paid}} \rangle.} \quad (2.6)$$

The fixed R-096 collar is a hypothesis used to bound (2.5a), not a second algebraic term. The separate R-086 Q orientations and $T_G^{\leq}$ current charge are not appended to (2.5a): their active pieces are reassigned by the R-101 raw-Wick coordinate and R-102 complete-current coordinate below. Formula (2.6) is simply the expectation of (2.3)–(2.5) after the lossless R-083 row and spatial projections.

3 The rational row has exactly three owners

At the heat-lifted rational endpoint set

$$\bar{B} = P_{\Sigma_{\text{tar}}} B_R, \quad B_0 = \bar{B}(U), \quad B_1 = \bar{B}(U + a), \quad Q = G^{\otimes 2} - \Gamma, \quad c = Da. \quad (3.1)$$

R-101 Theorem 6.1 proves the exact nonduplicating identity

$$\boxed{\Delta \mathcal{W}_R = \mathcal{R}_Q + \mathcal{M}_U + \mathcal{K}_R}, \quad (3.2)$$

where

$$\mathcal{R}_Q = \frac{1}{2}(B_1 - B_0) : Q, \quad (3.3)$$

$$\mathcal{M}_U = G^T B_0 c + G^T D \bar{B}(U)[a]c + \frac{1}{2} G^T D^2 \bar{B}(U)[a, a]c, \quad (3.4)$$

and $\mathcal{K}_R$ is (1.1), with

$$L = B_1 - B_0 - D \bar{B}(U)[a] - \frac{1}{2} D^2 \bar{B}(U)[a, a]. \quad (3.5)$$

Equation (3.2), rather than the old five-family split, is the active owner coordinate. Its three terms are assigned as follows.

1. **Raw-Wick residual owner** $\mathcal{R}_{Q,>}$. Apply R-101 Theorems 3.1 and 4.1 separately to the terminal and base coefficient processes $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(0)}$, whose endpoint difference is $B_1 - B_0$. If $F_j^{(s)} = \mathbb{E}_j \mathcal{B}^{(s)}(Z_*)$ for $s \in \{0, 1\}$, subtraction gives the exact expectation identity

$$\langle \mathcal{R}_Q \rangle = \underbrace{\frac{1}{2} \mathbb{E} \langle F_\ell^{(1)} - F_\ell^{(0)}, Q_\ell \rangle}_{\mathcal{R}_{Q,\text{low}}} + \underbrace{\frac{1}{2} \sum_{j>\ell} \mathbb{E} \langle [\mathcal{B}^{(1)}(Z_*) - \mathcal{B}^{(1)}(Z_j)] - [\mathcal{B}^{(0)}(Z_*) - \mathcal{B}^{(0)}(Z_j)], \Delta Q_j \rangle}_{\mathcal{R}_{Q,>}}. \quad (3.6)$$

The fixed-low difference $\mathcal{R}_{Q,\text{low}}$ is already inside the two R-080/R-063 low owners in (2.6); it receives no new row allowance. R-100 Theorem 5.1 pays only the future residual $\mathcal{R}_{Q,>}$ with one full-Wick R-094 secant allocation. Its covariance trace and complete R-063 forest are internal coordinates, not terms appended after (3.6). In particular, the old raw-Q Taylor families

$$\frac{1}{2} D \bar{B}(U)[a] : Q, \quad \frac{1}{4} D^2 \bar{B}(U)[a, a] : Q \quad (3.7)$$

must not be charged again.

2. **Unshifted current owner** $\mathcal{M}_U$. The three terms in (3.4), and only those three terms, are the accepted R-071/R-085 current families. They are the last three rows of the R-099 five-family table. Their individual exponents are strictly subcritical, so the proof of R-099 Theorem 7.1 applies to this subset without invoking the two forbidden raw-Q rows (3.7).

3. **Shifted current owner $\mathcal{K}_R$.** R-102 Theorem 9.2 and Corollary 9.3 prove

$$\mathbb{E}\mathcal{K}_R \geq -\eta\mathbb{E}X - \zeta\mathbb{E}Y - C_{\eta,\zeta,e,j_0} + \frac{1}{2}\mathbb{E}\sum_i \int c_i^T B_1 c_i dx, \quad (3.8)$$

in the declared regular class. Thus the entire terminal rational square is available as nonnegative reserve. The complete Hessian choice in R-102 requires refund of the separate provisional R-076 base-cubic allocation; the chronological coordinate also refunds the separate R-086 $T_G^{\leq}$ current charge. Neither predecessor theorem is withdrawn.

These assignments exhaust (3.2). The R-086 Q-oriented shifted diagnostics, a second R-094 secant, the R-076 base cubic, and the R-086 $T_G^{\leq}$ charge therefore have zero multiplicity in the active lower-bound ledger.

4 Complete-owner coordinate and posterior bookkeeping

The R-097 posterior bracket is

$$\mathfrak{C}_{\text{post}} = J_B + \mathbb{E}_{\mathcal{F}}[B : (V_B - \Gamma)] = \mathfrak{D}_R + \sum_i \int |r_i|^2 dx. \quad (4.1)$$

For clarity, the complete R-100 physical owner used below is

$$\mathcal{O}_{\mathcal{F}}(B, R) := S_R(B) + \frac{1}{2}\mathfrak{C}_{\text{post}}(B; R) - P_R(B, R) - W_0(B). \quad (4.1a)$$

The second term in the first expression has no standalone sign. R-100 Theorem 3.1 nevertheless proves exact row additivity of the physical complete owner. With matching row, payment, and baseline splits,

$$\mathfrak{C}_{\text{post}}(B; R) - \sum_a \mathfrak{C}_{\text{post}}(B_a; R_a) = D_{\text{row}}, \quad (4.2)$$

$$S_R(B) - \sum_a S_{R_a}(B_a) = -\frac{1}{2}D_{\text{row}}, \quad (4.3)$$

and hence

$$\boxed{\mathcal{O}_{\mathcal{F}}(B, R) = \sum_a \mathcal{O}_{\mathcal{F}}(B_a, R_a).} \quad (4.4)$$

For the production split into three linear rows and one rational row, (4.4) says that the Schur rebate, q/r conditional means, actual Cameron–Martin matrix payment, covariance trace, and posterior variance are internal coordinates of the row endpoints in (2.6). They are not additional terms. The positive cross-row Schur gap in (4.2) is cancelled exactly by (4.3); retaining both it and the complete endpoint would double count.

Theorem 4.1 (regular complete-owner partition)

On the declared regular class, (2.6) and (3.2) form a disjoint exact owner partition. Every term in the complete R-079 endpoint has multiplicity one, except the explicitly refunded provisional charges, which have multiplicity zero. The naked lower bound

$$\mathbb{E}\mathfrak{C}_{\text{post}} \geq -\eta\mathbb{E}X - \zeta\mathbb{E}Y - C \quad (4.5)$$

is not required for this partition and is not asserted.

Proof. Start from the exact R-079 identities (2.3)–(2.5). Apply the R-096 complete-cancellation-first ordering before the rational projection. The R-083 row identity partitions the production Gram into its three linear rows and one rational row, while its FAR/NEAR projections partition the spatial output. R-083 polynomial FAR vanishes. R-084 owns each complete linear-row endpoint. R-101 partitions the rational endpoint by (3.2). Equation (3.6) assigns the fixed-low raw-Wick difference to the already listed low owners and leaves exactly one future raw-Wick residual; (3.4) owns the three unshifted current families, and (3.8) owns the complete shifted current. R-080 keeps the two distinct low objects (2.3) and (2.5) once, while (2.5a) keeps the one R-078 paid difference, bounded with the declared R-096 collar. No second R-086 payment is present. Finally, (4.4) proves that changing from the endpoint coordinate to S_R , $\mathfrak{C}_{\text{post}}$, q/r , and payment coordinates has zero net remainder. The listed terms exhaust the exact partitions, proving the claim. $\square$

The theorem makes the full-frame posterior-covariance gate non-load-bearing for regular H_N/REG : its physical complete-owner target is supplied without the naked inequality. It does *not* close the gate’s literal stronger naked-bracket statement (4.5), which remains unproved and is recorded as superseded rather than proved.

5 Finite budget allocation closes complete regular H_N

Define the random variable $\mathfrak{H}_N^{\text{reg}}$ to be the complete NEAR packet whose expectation is the NEAR part of (2.6), with the two low objects and the one paid difference retained as required by R-091, R-096, and R-100. By Theorem 4.1 its negative contributions occur in the following finite list:

Owner	Authority	Active rule
Conditional low	R-080 Thm. 3.1	Sextic form bound; distinct from low endpoint.
Complete low endpoint	R-080 Thm. 3.2	Sextic form bound after paid subtraction.
Three linear rows	R-084 Thm. 7.1	Complete heat-lifted endpoints once.
Rational raw-Wick residual	R-100 Thm. 5.1; R-101 Thm. 4.1	Future residual only; fixed low already in the first two rows.
Rational unshifted current	R-071/R-085; R-099 (7.1)–(7.5)	Use only current families three through five.
Rational shifted current	R-102 Thm. 9.2, Cor. 9.3	Keep terminal square unspent.
Paid difference	R-078/R-096	The single signed difference (2.5a); fixed collar is a bound hypothesis.

Theorem 5.1 (complete regular H_N closure)

For every $\eta, \zeta > 0$, (1.2) holds uniformly in the terminal cutoff and deterministic positive-semidefinite target/future heat.

Proof. Give each of the seven rows in the table the internal allowances $\eta/7$ and $\zeta/7$; rows using only one budget simply leave the other allowance unused. R-080 proves the first two bounds. R-084 proves the third. R-100/R-101 prove the fourth after the complete raw-Wick telescope. The termwise inequalities underlying R-099 Theorem 7.1 prove the fifth for the three current families. R-102 proves the sixth, and the last row is the accepted R-078/R-096 payable remainder. Each bound is uniform in the terminal cutoff in the stated common scope. The sum of the seven allowances is at most (η, ζ) . The exact partition of Theorem 4.1 proves (1.2). $\square$

The proof uses a finite allocation after all root and shell sums have been performed inside their authorities. It does not apply Young’s inequality root by root, and it creates no new moment requirement. The limiting moment remains the R-102 high-prefix moment 14; its fixed-low branch requires only 42/17, while older complete-linear/current models have smaller accepted requirements. For the conservative

choice (1.5), the seven NEAR shares are $1/3080$ and $3/700$, while the eight REG shares are $1/3520$ and $3/800$.

6 Cartan FAR and REG

The exact R-083 stopped-current theorem kills every controlled polynomial FAR row for the declared collar. For the Cartan row, R-092 proves

$$\mathcal{H}_C^{\text{reg}}(C) \leq 2^{-(C-5)/2} \mathcal{B}_{1/4}^{\text{out}}, \quad (6.1)$$

and

$$\mathcal{B}_{1/4}^{\text{out}} \leq C_{e,L} \{C_{\epsilon,\delta,e,L} + \epsilon \mathbb{E}X + \delta \mathbb{E}Y\}. \quad (6.2)$$

The actual signed Cartan FAR contribution carries only its fixed production prefactor, so (6.1)–(6.2) give an arbitrary X/Y form allocation. For example $C = 15$ makes the explicit gap in (6.1) equal to $2^{-5} = 1/32$; no special value of C is needed for the theorem.

Theorem 6.1 (regular REG closure)

For every $\eta, \zeta > 0$,

$$\boxed{\mathbb{E}\mathfrak{P}_{\text{reg}} \geq -\eta \mathbb{E}X - \zeta \mathbb{E}Y - C_{\eta,\zeta,e,j_0,C}.} \quad (6.3)$$

Consequently (1.4) holds with the explicit choice (1.5).

Proof. Use Theorem 5.1 for complete NEAR and its low/paid companions. Add the zero R-083 polynomial FAR term and (6.1)–(6.2) for Cartan FAR. Equivalently, allocate $(\eta/8, \zeta/8)$ to the eight modules consisting of the seven rows in Section 5 and the Cartan FAR module. Their exact sum is (2.6), so (6.3) follows. Choose $\eta = \eta_* = 1/440$ and $\zeta = \zeta_* = 3/100$ and add the stabilizers in (1.3). Since

$$c_X - \eta_* = \frac{197}{440} > 0, \quad c_Y - \zeta_* = \frac{3}{25} > 0, \quad (6.4)$$

equation (1.4) follows. $\square$

This is REG in its intended sense: a cutoff-uniform lower bound on the complete packet for the regular graph class. It is a sufficient proof architecture, not part of the R-087 CORE equivalence and not a theorem for all progressive controls.

7 Ownership audit and forbidden duplicate routes

The complete multiplicity ledger is

quantity	active owner count	reason
two low objects	1 each	R-080 distinguishes them
linear row heat/trace/forest	1	inside R-084 endpoint
future raw-Wick residual/trace/forest	1	low part already in R-080 rows
R-094 full-Wick secant	1	inside R-100 residual
q/r and Schur rebate	0 extra	coordinate identity (4.4)
R-078 paid difference	1	R-096 collar bounds the same term
terminal rational square	1 positive	reserved by R-102
R-076 base cubic	0	refunded for complete L
R-086 $T_G^{\leq}$ charge	0	refunded in chronological coordinate
R-086 separate Q orientations	0	inside low/raw-Wick split (3.6)
old raw-Q Taylor charges	0	contained in raw-Wick owner

(7.1)

The following tempting conclusions remain invalid.

1. A lower bound for the naked $\mathfrak{C}_{\text{post}}$ is not obtained by discarding the complete Schur square. The abstract R-097/R-100 negative fixtures remain valid method boundaries.
2. R-085 (4.11) is not recovered by summing pathwise translated atoms. Regular Cartan FAR is closed by the integrated normalized-lift theorem R-092.
3. Old R-085 (6.5) is not recovered by separately paying $L : Q/2$ and $G^T Lc$. The first lies in (3.6); the second lies in (3.8).
4. REG does not imply H_A or $\text{OVERLAP}_{\text{src}}$. A revisit can have small or zero terminal displacement and a large intermediate prefix. The R-080/R-091/R-094 counterexamples remain outside the regular class and still block this inference.

8 Attempt, failure, and evidence map

The successful proof path is

$$\begin{array}{c}
\text{R-079 exact endpoint and safe-packet reconstruction} \\
\downarrow \\
\text{R-080 two low owners; R-083 exact FAR/NEAR and row partition} \\
\downarrow \\
\text{R-092 regular } H_C \quad + \quad \text{R-084 complete linear NEAR} \\
\downarrow \\
\text{R-096 cancellation first; R-100/R-101 raw-Wick ownership} \\
\downarrow \\
\text{R-101 } \Delta \mathcal{W}_R = \mathcal{R}_Q + \mathcal{M}_U + \mathcal{K}_R \\
\downarrow \\
\text{R-071/R-085 unshifted current} \quad + \quad \text{R-102 } \mathcal{K}_R \\
\downarrow \\
\boxed{\text{R-103 complete regular } H_N \text{ and REG}} \\
\downarrow \\
\boxed{\text{next: progressive/revisit } H_A \text{ and } \text{OVERLAP}_{\text{src}}}.
\end{array} \tag{8.1}$$

The reusable route decisions are recorded in `explorations/log.jsonl`. They include the successful exact-owner reassembly and the boundary that standalone posterior positivity and the old R-085 atom statements are unnecessary, not retroactively proved. No new negative-result entry is required: the relevant duplicate-owner and regular-to-progressive failures were already registered by R-091, R-094–R-101.

9 Devil’s-advocate review

1. **Objection: the sign-indefinite posterior covariance deficit is still open. DISMISSED for regular complete-owner closure; VALID for the stronger naked bracket.** Equations (4.2)–(4.4) show that its rowwise Schur gap cancels the complete-square gap. The physical endpoint, not the naked bracket, is the target. No positivity of (4.5) is claimed.
2. **Objection: R-099 already spent all five rational families, so R-101 spends the first two again. DISMISSED by the active ledger.** R-103 uses only R-099 families three through five. Families one and two are inside the single R-101 raw-Wick owner (3.6).

3. **Objection: the R-102 complete Hessian and R-076 cubic are both needed. UPHeld as a duplicate-charge warning.** The active proof refunds the provisional R-076 cubic before taking the complete Hessian. R-076 remains valid in its complementary coordinate.
4. **Objection: q , r , trace, heat, and the forest are absent from (2.6). DISMISSED.** They are internal coordinates of the complete row endpoints. Equations (3.6) and (4.4) show why appending them would be a second copy, not a repair.
5. **Objection: the terminal square is secretly used by R-100 or R-102. DISMISSED.** R-100 pays only the signed future raw-Wick residual. R-102 proves an absolute current estimate and Corollary 9.3 leaves the entire square as the explicit positive term in (3.8).
6. **Objection: regular REG plus R-087 CORE proves Nelson. UPHeld as false.** CORE identifies variational classes only after a uniform bound on bounded smooth cylindrical-simple progressive controls. The missing theorem is the lossless revisit/overlap-stable H_A assembly and its uniform source-action lower bound.
7. **Objection: deterministic PSD heat covers control-dependent heat. UPHeld as false.** R-102 conditions through deterministic heat composition. Random control-dependent target heat is outside the theorem.

External adversarial review is invited for the exact equality (2.6), the three-owner rational split (3.2), the zero-extra-owner interpretation of q/r in (4.4), the refund rows in (7.1), and the regular/progressive scope boundary.

10 Executable verification

The primary executable checks the exact rational endpoint algebra on deterministic exact rational matrix fixtures, rejects wrong raw-Wick, shifted-current-sign, and terminal-square factors, and includes rank-one and zero PSD terminal matrices. It also checks the owner-incidence partition, active/refund multiplicities, seven- and eight-module budget simplices, production stabilizer reserve, the $C = 15$ R-092 Cartan separation gap (distinct from the R-096 payable collar), and predecessor contracts. The independent executable imports no primary implementation; it uses standard-library **Fraction** arithmetic, a separately expanded scalar endpoint including the zero-PSD limit and mutant rejection, and an independently ordered owner incidence audit.

The integrated verifier runs both executables from fresh outputs, pins every authority and evidence hash, checks the note and PDF, verifies the exploration records, result ledger, current claim surface, gate and task states, and fails closed on any promotion to progression, Nelson, removals, a measure, Sector A, or a higher tier.

Run from the repository root:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_regular_complete_packet_ownership_hn_reg_closure_verify.py
```

The PDF is built from this source with the repository PDF builder and rendered page by page with Poppler before release.

Result footer

Result ID: A13-CLASSII-REGULAR-COMPLETE-PACKET-OWNERSHIP-HN-REG-CLOSURE
 Ledger: R-103.
 Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
 Precise statement:

The R-079 regular complete packet has one exact disjoint owner partition. R-080 owns its two low objects, R-083/R-092 own FAR, R-084 owns the complete linear rows, and R-101 splits the rational row into a low-owned raw-Wick part, one future raw-Wick residual, three unshifted currents, and K_R. R-100/R-101, R-071/R-085, and R-102 bound those three rational owners, while q/r, heat, trace, payment, and the R-063 forest remain internal coordinates. Seven-module and eight-module finite allocations close complete regular H_N and REG. The conservative allocations 1/440 and 3/100 retain positive action reserves 197/440 and 3/25; 1/220 is only an optional R-088 comparison threshold.

Scope: Finite cutoff, fixed positive floor, A1 production frame, common real-even local regulators, deterministic PSD target/future heat, fixed R-096 collar, and annular mutually orthogonal strict-past no-revisit physical-shell controls with accepted prefix ledgers.

Dependencies:

R-063, R-071, R-076, R-078--R-080, R-083--R-086, R-088, R-091--R-094, R-096--R-097, R-099--R-102.

Evidence grade:

ANALYTIC, EXACT, EXECUTED, REGULAR-CLOSURE.

Reproduction command:

E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_regular_complete_packet_ownership_hn_reg_closure_verify.py

Expected output:

Manifest-pinned primary, independent, integrated, and aggregate assertions PASS; exit zero.

Falsification gate:

Failure of the R-079 endpoint identity, R-083 partition, R-100 row additivity, R-101 rational split/raw-Wick ownership, R-102 K_R bound, any owner multiplicity, finite allocation, authority hash, PDF, exploration, public-surface, or scope assertion.

Tier before / after:

T4 / T4; no promotion.

No-overclaim statement:

No naked posterior-bracket positivity, standalone R-085 (4.11) or old (6.5), arbitrary progressive/revisit H_A, OVERLAP_src, Nelson, cutoff/floor removal, interacting measure, T5--T7, or Sector-A closure is proved.

Next required action:

Prove lossless progressive/revisit H_A assembly and the uniform OVERLAP_src lower bound on the R-087 bounded smooth cylindrical-simple variational core, then invoke CORE for q=10/9 Nelson.